



**Software Engineering Department
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Maintenance Guide

Characterizing Normal Brain Development via EEG Pattern Matching

CODE: 26-1-R-11

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1. Purpose and Maintenance Scope

This maintenance guide is intended for future developers, students, or researchers who need to continue using, modifying, or extending the EEG Motif Discovery Pipeline after the project submission. The guide focuses on the maintainable structure of the Google Colab notebook, the cached output folders, the important configuration parameters, and the correct rerun strategy after code or parameter changes.

The system is a research pipeline rather than a conventional end-user application. Therefore, maintenance mainly means keeping the notebook reproducible, keeping the dataset paths and task names correct, updating parameters safely, and regenerating outputs without repeating expensive preprocessing unnecessarily.

- Intended maintainer: a developer familiar with Python, Google Colab, Google Drive, EEG data, and basic signal processing.
- Maintenance focus: environment setup, parameter changes, task/channel updates, output regeneration, validation, and documentation consistency.
- Out of scope: redesigning the research question, manual editing of generated result files, or changing the HBN-EEG dataset itself.

2. System Overview

The project characterizes normal brain development by discovering recurring EEG window patterns, called motifs, across age groups and channels. The pipeline loads HBN-EEG recordings, preprocesses them, splits the cleaned signal into windows, encodes every window using CTM and OPM, builds motif libraries, and generates statistics and plots.

The main design decision is to compare signal shape rather than raw amplitude. EEG amplitudes can vary strongly across participants, so the pipeline uses shape-based encodings: OPM captures local rank order, while CTM captures global peak-and-valley structure. This makes the outputs more interpretable for age-group comparison.

Stage	Purpose	Main outputs
Stage 1 - Preprocess	Load EEGLAB .set files, filter the signal, remove artifacts using ICA + ICLabel, select channels, and apply z-score normalization.	01_cleaned/ files containing cleaned signals, sampling rate, selected channel names, and preprocessing metadata.
Stage 2 - Window + Encode	Slice cleaned signals into fixed-length windows and encode each window with CTM and OPM while preserving channel indices.	02_codes_L{L}/ files containing windows, ctm_codes, opm_codes, channel indices, and ch_names.
Stage 3 - Motif Libraries	Count code presence per subject and group, then select motifs according to the IN/OUT rule.	03_libraries_L{L}/ pickle files with per-task, per-group, per-channel motif libraries.
Stage 4 - Analysis and Plots	Run statistical tests, diversity plots, motif-shape plots, and motif-frequency plots.	04_results_L{L}/ figures and summary outputs for the selected L.

3. Environment and Required Software

The pipeline is implemented as a Google Colab notebook and uses Google Drive for both the HBN-EEG dataset and cached outputs. A High-RAM runtime is recommended when processing many participants or scanning several window lengths.

Component	Required / Recommended
Runtime	Google Colab, preferably High-RAM for full runs.
Storage	Google Drive access to the shared project folder and enough storage for cleaned signals, encoded windows, libraries, and plots.
Dataset	HBN-EEG dataset in BIDS format, folder name ds005505.
Core packages	mne, mne-icalabel, numpy, pandas, scipy, matplotlib. The notebook installs the main EEG packages in Cell 1.
Input EEG format	EEGLAB .set files inside subject/session/eeg folders.

Output format	.npz files for cached arrays, .pkl for motif libraries, image files for plots.
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Installation note: The maintainer does not need a separate desktop installation. Open the notebook in Google Colab, connect Drive, run Cell 1, and allow the notebook to install or import the required Python packages. If the Colab session is reset, rerun Cell 1 before continuing.

4. Dataset, Repository, and Folder Structure

The notebook expects a BIDS-style dataset root and a separate working root for outputs. In the supplied code, these are configured in Cell 1 using `base_path` and `WORK_ROOT`.

Path / variable	Role
<code>base_path</code>	Root path to the ds005505 HBN-EEG dataset in Google Drive. This folder must contain <code>participants.tsv</code> and <code>sub-*</code> folders.
<code>WORK_ROOT</code>	Root path for all generated pipeline outputs. The notebook creates the stage folders below it.
<code>CLEANED_DIR</code>	Folder for Stage 1 cleaned files. It is usually <code>WORK_ROOT/01_cleaned</code> .
<code>dirs_for_L(L)</code>	Utility function that creates/returns <code>02_codes_L{L}</code> , <code>03_libraries_L{L}</code> , and <code>04_results_L{L}</code> .

Expected dataset structure:

- `ds005505/participants.tsv`
- `ds005505/sub-<subject_id>/ses-<session_id>/eeg/sub-<subject_id>_task-<TaskName>_eeg.set`
- Optional metadata files: `*_events.tsv`, `*_channels.tsv`, `*_eeg.json`

The file discovery logic should be kept flexible because not every participant has every task or run. Missing task files should be skipped, not treated as a full pipeline failure.

5. Notebook Architecture and Cell Responsibilities

The current notebook is organized into 12 functional cells. Maintainers should avoid merging unrelated stages because the staged structure supports debugging and efficient reruns.

Cell	Name	Maintainer meaning	Rerun when
Cell 1	Setup	Install/import packages, mount Drive, define paths, task list, channel list, preprocessing parameters, motif thresholds, window grid, and output folders.	After changing paths, tasks, channels, thresholds, window grid, or preprocessing defaults.
Cell 2	Optional cleanup	Delete Stage 2 and Stage 3 cached outputs while keeping Stage 1 cleaned signals.	When rebuilding encoding/libraries but not ICA.
Cell 3	Participants and age groups	Load <code>participants.tsv</code> , keep ages 5-21, split subjects into three quantile-based groups, and prepare the subject pool.	After changing age filtering, grouping logic, or scan/full mode.
Cell 4	CTM encoder	Define CTM parent-distance encoding and run a unit test.	After editing CTM logic.
Cell 5	OPM encoder	Define OPM natural rank representation and run a unit test.	After editing OPM logic or tie handling.
Cell 6	Stage 1 preprocess	Load <code>.set</code> files, filter, run ICA + <code>ICLabel</code> , select channels, normalize, and save to <code>01_cleaned/</code> .	After changing preprocessing settings or channel list.
Cell 7	Stage 2 encode	Create windows for each L and save CTM/OPM codes with channel tracking into <code>02_codes_L{L}/</code> .	After changing L, overlap, duration limit, or cleaned inputs.
Cell 8	Stage 3 motif library	Build motif libraries for one selected L using the IN/OUT rule and <code>MIN_OCC_PER_SUBJECT</code> .	After changing motif strictness or rerunning Stage 2.
Cell 8.5	Grid search	Scan multiple L values and summarize motif counts to help choose the window representation length.	When tuning L or comparing parameter configurations.
Cell 9	Statistical tests	Run per-channel Kruskal-Wallis tests for selected metrics and <code>L_FOR_STATS</code> .	After selecting a final L or regenerating codes.
Cell 10	Per-channel plots	Generate diversity and motif-shape plots for <code>L_FOR_PLOTS</code> .	After rebuilding libraries or changing plot targets.
Cell 11	Motif mean +/- std and consistency checks	Generate optional motif-shape plots and verify window-length consistency.	Before reporting final motif-shape results.

Cell 12	Motif frequency	Plot median motif-frequency percentage per channel across subjects for the selected task and L.	For final representative frequency figures.
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6. Key Configuration Parameters

Most maintenance changes should be made in Cell 1. Keeping the main parameters centralized reduces mismatches between preprocessing, encoding, motif discovery, and plotting.

Parameter	Meaning	Maintenance guidance
base_path	Google Drive dataset root. Must point to ds005505.	Change only when dataset location changes.
WORK_ROOT	Google Drive output root.	Changing it creates a new cache/output tree.
RUN_MODE	Controls whether the notebook uses the full subject pool or a scan subset.	Use scan mode for quick tests; use full mode for final outputs.
SCAN_PER_GROUP	Number of subjects per group for scan mode.	Keep small during tuning to reduce runtime.
TASKS	List of task names to process.	Must match BIDS file task names exactly. Add DespicableMe, RestingState, or contrastChangeDetection_run-2 only if those files are available.
CHANNELS_45 / FRONTAL_CHANNELS	Selected EGI electrode labels. The variable name is kept for compatibility with older cells.	If channels change, rerun Stage 1 onward. Verify the printed channel count because the supplied list may contain more than 45 unique channels.
FILTER_L / FILTER_H	Bandpass filter limits.	If changed, clear/rebuild Stage 1 because cleaned signals change.
ICA_N_COMPONENTS / ICA_RANDOM_STATE	ICA artifact-removal settings.	If changed, rebuild Stage 1 and document the new configuration.
OVERLAP	Window overlap ratio.	If changed, rebuild Stage 2 onward.
WINDOW_LENGTHS_GRID	L values scanned in Cell 8.5.	Use a small grid first. The current code uses [8, 10].
L_FOR_STATS / L_FOR_PLOTS / L_FOR_FREQ	Selected L for statistical tests, motif-shape plots, and frequency plots.	Keep these synchronized when preparing final figures.
IN_GROUP_PCT	Minimum fraction of target-group subjects that must contain a code for it to become a motif.	If changed, rebuild Stage 3 onward.
OUT_GROUP_PCT	Maximum fraction allowed in other groups. In the supplied code, the real stored-library value is used for filenames.	Make sure graph titles and report text match the real value used in the library.
MIN_OCC_PER_SUBJECT	Minimum number of times a code must occur in a subject to count as present.	If changed, rebuild Stage 3 onward.
FORCE_STAGE1	Forces rebuilding cleaned files.	Set True once after channel/preprocessing changes; set back to False after Stage 1 finishes.
MAX_SECONDS_PER_SUBJECT	Drive-safe duration limit per subject.	Increase only after confirming the pipeline is stable.

Important consistency note: The current notebook contains display-only labels in some plotting cells that can show $OUT \leq 20\%$ even when the real `OUT_GROUP_PCT` used for the library is different. Before final reporting, either align `OUT_GROUP_PCT` with the displayed rule or update the plot title code so that the displayed rule equals the real library rule.

7. Output Folders and Caching Strategy

Caching is central to this project because ICA preprocessing is slow and Google Drive I/O can be unstable during large runs. Maintainers should regenerate only the stages affected by the change.

Folder	Created by	Contains	Rebuild when
01_cleaned/	Cell 6	Cleaned signals after filtering, ICA + ICLabel,	Preprocessing settings, channel

		channel selection, and z-score normalization.	list, task list, or raw data path changed.
02_codes_L{L}/	Cell 7 or Cell 8.5	Window arrays, CTM/OPM codes, channel indices, and channel names for a specific L.	L, overlap, MAX_SECONDS_PER_SUBJECT, or cleaned inputs changed.
03_libraries_L{L}/	Cell 8 or Cell 8.5	Motif libraries per task, group, channel, and encoding type.	IN/OUT rule, MIN_OCC, group split, or Stage 2 files changed.
04_results_L{L}/	Cells 9-12	Statistical results, diversity plots, motif-shape plots, frequency plots, and validation figures.	Whenever new plots or updated figures are needed.

Do not manually rename or edit cached files unless you are intentionally archiving a run. The naming convention encodes task, L, and motif rule, and later cells depend on those names.

8. Standard Maintenance Workflows

Maintenance change	Minimum rerun path	Notes
Change only plot style or titles	Cell 10/11/12	No need to rerun preprocessing, encoding, or motif libraries.
Change task list	Cell 1 -> Cell 6 -> Cell 7 -> Cell 8 -> Cells 9-12	Task names must match files exactly. Missing tasks should be skipped by discovery logic.
Add or remove channels	Cell 1 -> Cell 6 -> Cell 7 -> Cell 8 -> Cells 9-12	Stage 1 must be rebuilt because channel selection happens during preprocessing.
Change window length L grid	Cell 1 -> Cell 7 or 8.5 -> Cell 8 -> Cells 9-12	Stage 1 can be reused if preprocessing and channels did not change.
Change IN/OUT motif rule	Cell 1 -> Cell 8 -> Cells 9-12	Do not rerun ICA or Stage 2; rebuild motif libraries and plots.
Change age grouping logic	Cell 3 -> Cell 8 -> Cells 9-12; rerun Stage 2 only if subject pool changed and missing encoded files are needed	Update documentation and poster labels if group definitions change.
Change preprocessing filter or ICA settings	Cell 1 -> Cell 6 -> Cell 7 -> Cell 8 -> Cells 9-12	This changes the underlying cleaned signal, so all later outputs become outdated.
Drive disconnects during Stage 2	Rerun Cell 7 for the affected L/task after remounting Drive	The code saves locally first and then copies to Drive; keep this pattern.

8.1 Recommended Final-Run Procedure

1. Confirm base_path and WORK_ROOT in Cell 1.
2. Set TASKS to the final tasks used in the report/poster.
3. Confirm channel list and print the channel count.
4. Run Cell 3 and check that the age groups contain the expected number of participants.
5. Run Cells 4 and 5 and confirm both encoder unit tests pass.
6. Run Cell 6 only if Stage 1 must be rebuilt; otherwise reuse 01_cleaned.
7. Run Cell 7 or Cell 8.5 for the selected L values.
8. Run Cell 8 to build motif libraries for the final L and rule.
9. Run Cells 9-12 to produce statistics, diversity plots, motif-shape plots, and motif-frequency plots.
10. Copy final outputs to the report/poster only after confirming titles, task names, channels, and IN/OUT labels match the actual run.

9. Validation and Quality Checklist

Use this checklist before submitting results or changing the report/poster.

- Cell 1 prints the expected task list, channel count, L grid, and force flags.
- Cell 3 prints the expected number of subjects per group and the intended age range 5-21.
- CTM and OPM unit tests pass in Cells 4 and 5.
- Stage 1 created or reused cleaned files for every available subject-task pair.
- Stage 2 output files exist for the selected L in 02_codes_L{L}/.
- Stage 3 library files exist and their filenames match the real IN/OUT rule.

- Plot titles show the correct task, channel, L, and motif rule.
- Motif-shape plots use the same L as the motif library and encoded files.
- Motif-frequency plots show only channels where at least one group produced qualifying motifs.
- Report, poster, and notebook agree on the task names, number of groups, channel list, and final parameters.

9.1 Code-Level Validation Tests

Validation	Where	Expected result
CTM encoder unit test	Cell 4	The printed result should confirm the computed parent-distance code matches the expected test output.
OPM encoder unit test	Cell 5	The printed result should confirm the natural rank representation is correct.
Window-length consistency check	Cell 11	All motif windows used for plotting should have the same L.
Kruskal-Wallis tests	Cell 9	Statistics should run without empty-group errors; reported p-values should be linked to the selected metric and L.
Frequency plot verification	Cell 12	Bars should be grouped by channel, use correct group colors, and keep E-number channel labels.

10. Troubleshooting Notes

Problem	Likely cause	Recommended action
Transport endpoint is not connected / Drive I/O error	Google Drive disconnected or too many files written during a long stage.	Remount Drive, keep MAX_SECONDS_PER_SUBJECT low during scans, run fewer L values, and keep the local-save-then-copy pattern.
File not found for a task	A participant does not contain that task or run naming differs.	Check BIDS filenames and use robust file discovery. The pipeline should skip unavailable subject-task files.
No motifs found for a group or channel	The IN/OUT rule may be too strict, L too long, or the group too small.	Use Cell 8.5 to scan L and consider adjusting IN/OUT or MIN_OCC only after documenting the change.
Too many motifs or weak separation	The rule may be too loose or L too short.	Increase strictness or use a different L; compare motif frequency and diversity outputs.
Mismatch between graph title and library filename	Display-only labels in plot cells differ from real IN/OUT settings.	Make titles read from IN_GROUP_PCT and OUT_GROUP_PCT, or set those values to the displayed rule before final runs.
Poster/report task list differs from notebook	TASKS changed during experimentation but documentation was not updated.	Update Dataset/Task Context sections and rerun the final plots for the listed tasks.
Channel count mismatch	The variable name CHANNELS_45 may not match the actual number of unique channels.	Run len(set(CHANNELS_45)) and update the poster/report text accordingly.

11. Documentation and Reporting Consistency

The project passed through multiple parameter experiments: different task selections, different age-group splits, several motif rules, and different window lengths. Because of that, the maintainer must keep the code, user guide, maintenance guide, report, poster, and final plots synchronized.

Item to synchronize	Where it appears	How to verify
Team code and project title	Cover pages, poster, GitHub README	Use CODE: 26-1-R-11 and the same project title everywhere.
Tasks	TASKS variable, report Task Context, poster Dataset section, plot titles	Check plot titles and file names in 04_results_L{L}/.
Age groups	Cell 3, report dataset split, poster explanations	Record whether groups are quantile-based and how many subjects per group are used.
Channels	CHANNELS_45 / FRONTAL_CHANNELS,	Print actual channel count and use the

	poster channel map, report method section	same E-number list in documentation.
Window length L	WINDOW_LENGTHS_GRID, L_FOR_STATS, L_FOR_PLOTS, L_FOR_FREQ, plot titles	Confirm all final figures use the intended L.
IN/OUT rule	IN_GROUP_PCT, OUT_GROUP_PCT, library filenames, plot titles, report captions	Make sure displayed percentages match the real library files.
References	Report, guides, presentation	Keep the same core references for HBN-EEG, OPM, CTM, and EEG preprocessing.

12. References

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